

Sabancı University

DSA210 Project Final Report

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1. Project Motivation

This project investigates whether directors experience a "fatigue effect" after making long movies. The main research question is that when a director makes a long movie (over 150 minutes), does it negatively affect the rating of his/her next movie?

2. Data Source

2.1 IMDb Dataset

- Source: Kaggle
- Name: IMDb Top 1000 Movies
- Content: 1000 movies, 16 columns
- Used columns: Series_Title, Director, Runtime, IMDB_Rating, Released_Year

2.2 TMDb Dataset

- Source: Kaggle
- Name: TMDb 5000 Movies Dataset
- Content: 5000 movies, 20 columns
- Used columns: original_title, runtime, popularity

2.3 Collection Method

- Both datasets were downloaded as CSV files from Kaggle.
- No API or web scraping was required.
- Data was loaded into Python using pandas and merged on movie titles.

3. Data Analysis

3.1 Data Cleaning and Merging

- Merged IMDb and TMDb on movie title (Series_Title = original_title) using inner join
- Result: 459 common movies
- Converted Released_Year to numeric format
- Removed rows with missing director or year information

3.2 Exploratory Data Analysis (EDA)

- Histograms for runtime, rating, and year distributions
- Correlation matrix between numerical variables
- Identified that most movies are between 90-120 minutes

3.3 Data Preparation for Hypothesis Testing

- Sorted movies by director and year
- Added previous movie information using shift(1) within each director group
- Calculated rating difference: rating_diff = current_rating - previous_rating
- Removed first movies (no previous movie to compare)

3.4 Hypothesis Testing (t-test)

Null Hypothesis (H_0): There is no difference in rating change between directors whose previous movie was long (>150 min) and those whose previous movie was short (≤ 150 min).

Alternative Hypothesis (H_1): Directors who made a long movie experience a greater drop in their next movie's rating.

Groups:

- Long previous movie: $\text{prev_runtime} > 150$ (39 movies)
- Short previous movie: $\text{prev_runtime} \leq 150$ (145 movies)

Method: Independent two-sample t-test

3.5 Machine Learning

Two models were trained to predict rating_diff using prev_runtime and prev_rating :

Model	R^2 Score	RMSE
Linear Regression	0.395	0.273
Random Forest	0.088	0.335

Linear Regression performed better, suggesting a linear relationship.

4. Findings

4.1 Hypothesis Test Results

Group	Number of Movies	Average Rating Change
Long previous movie (>150 min)	39	-0.205
Short previous movie (≤ 150 min)	145	-0.023

p-value = 0.0106 (< 0.05)

The difference is **statistically significant**

4.2 Interpretation

Directors who make a movie longer than 150 minutes see their next film's rating drop by **0.205 points on average**. Directors with shorter previous movies experience almost no change (-0.023).

The hypothesis is supported.

4.3 Machine Learning Results

- Linear Regression achieved $R^2 = 0.395$, meaning previous movie runtime and rating explain about 40% of the variation in rating change.
- Random Forest performed worse, indicating the relationship is more linear than complex.

5. Limitations and Future Work

5.1 Limitations

Limitation	Explanation
Sample size	Only 459 movies could be matched between IMDb and TMDB
Indirect measurement	"Fatigue" is measured via runtime, not directly observed
Missing variables	Budget, cast size, genre, and production company were not included
Top-rated bias	IMDb Top 1000 only includes high-rated movies; results may not generalize to all films

5.2 Future Work

- Include more features: budget, cast size, genre, production company
- Analyze specific director career patterns (e.g., Nolan, Scorsese, Tarantino individually)
- Apply time series analysis to detect recovery periods
- Test with non-English cinema datasets
- Use sentiment analysis on user reviews as an alternative quality metric

6. Conclusion

This project successfully demonstrated that:

- Directors pay a small but measurable price for making long movies (≈ 0.2 point drop on IMDb's 10-point scale).
- The effect is **statistically significant** ($p = 0.0106$).
- A linear regression model using previous movie features can predict rating change with $R^2 = 0.395$.

The director fatigue hypothesis is **supported** by the data.

7. References

- IMDb Top 1000 Movies Dataset – Kaggle
<https://www.kaggle.com/datasets/harshitshankhdhar/imdb-dataset-of-top-1000-movies-and-tv-shows>
- TMDb 5000 Movies Dataset – Kaggle
<https://www.kaggle.com/datasets/tmdb/tmdb-movie-metadata>
- Python Libraries: pandas, matplotlib, seaborn, scipy, scikit-learn
- Project Guidelines: DSA210 Spring 2026